STANDATA interpretation 23-BCI-030/23-FCI-012

Building/Fire

Measures to mitigate fire spread to adjacent buildings

Date Issued: March 2025

Purpose

To clarify the requirements for measures to mitigate fire spread to adjacent buildings, water supply, and access for firefighting as per the National Building Code – 2023 Alberta Edition (NBC(AE)) and the National Fire Code – 2023 Alberta Edition (NFC(AE)).

Discussion

The NBC(AE) has specific provisions for firefighting under Subsection 3.2.5. such as fire access routes and water supply requirements. Additionally, Part 8 addresses safety measures at construction and demolition sites and references the NFC(AE).

In the NFC(AE), Section 5.6., “Construction and Demolition Sites,” and Section 2.8., “Emergency Planning,” specify that fire safety plans are mandatory requirements to be prepared in cooperation with the fire department and the authority having jurisdiction (AHJ), such as a building safety codes officer (building SCO) or fire safety codes officer (fire SCO). In municipal areas that are unaccredited in the fire discipline, fire safety plans must be accepted in writing by the building SCO and the fire department. In municipal areas that are accredited in the building and fire disciplines, these plans are reviewed by both disciplines; however, it is typically the fire SCO that provides the acceptance of the final plans.

Prior to the construction, alteration, or demolition of a building, the owner, developer, or contractor who is responsible for a construction or demolition site is required to submit the mandatory fire safety plan to the AHJ. The measures outlined in the fire safety plan should provide a reasonable reassurance that fire hazards are removed and/or controlled on the site, so that fires that originate in a building or facility undergoing construction or demolition will not quickly spread to exposed adjacent buildings or facilities (NFC(AE) Article 5.6.1.2.). These measures must consider the size, type, and configuration of the project, the methods of construction or demolition to be utilized and the distance to, and stage of completion of, adjacent exposures. Solutions may be active, passive, or a combination of active and passive, to achieve the desired protection. As such, there is no “one solution” that will be appropriate for all projects.

Furthermore, the owner, developer, or contractor responsible for the site is required to ensure that the fire department has unobstructed access to the site and that water supply for firefighting operations is available and sufficient to control fire situations (NFC(AE) Article 5.6.1.4.).

This approval is for all construction, alteration, or demolition operations.

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Code References

NBC(AE) and NFC(AE) Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms

Alteration means a change or extension to any matter or thing or to any occupancy regulated by this Code.

Authority having jurisdiction means a safety codes officer in the fire discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the *Safety Codes Act*.

Building means any structure used or intended for supporting or sheltering any use or occupancy.

Combustible construction means that type of construction that does not meet the requirements for noncombustible construction or encapsulated mass timber construction.

Supervisory staff means those occupants of a building who have some delegated responsibility for the fire safety of other occupants under the fire safety plan.

A-1.4.1.2.(1) Authority Having Jurisdiction

The defined term “authority having jurisdiction” refers to a safety codes officer as the appointed individual responsible for decision-making in a number of areas. In most cases, this individual will be a member of an accredited municipal fire department. In municipalities that use accredited agencies for their inspection and enforcement, the fire department may not be involved. It is important that accredited municipalities employing accredited agencies properly designate the person(s) acting as the authority having jurisdiction.

In addition, municipalities using accredited agencies should ensure that the agency has a close working relationship with the fire department during the decision-making process.

NBC(AE) Article 3.2.5.6.

3.2.5.6. Access Route Design

1) A portion of a roadway or yard provided as a required access route for fire department use shall

a) have a clear width not less than 6 m, unless it can be shown that lesser widths are satisfactory,
b) have a centre-line radius not less than 12 m,
c) have an overhead clearance not less than 5 m,
d) have a change of gradient not more than 1 in 12.5 over a minimum distance of 15 m,
e) be designed to support the expected loads imposed by firefighting equipment and be surfaced with concrete, asphalt or other material designed to permit accessibility under all climatic conditions,
f) have turnaround facilities for any dead-end portion of the access route more than 90 m long, and
g) be connected with a public thoroughfare.

(See Note A-3.2.5.6.(1).)

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2) For *buildings* conforming to Article 3.2.2.51. or 3.2.2.60., no portion of the access route described in Sentence 3.2.2.10.(3) shall be more than 20 m below the uppermost floor level.

A-3.2.5.6.(1) Fire Department Access Route. The design and construction of fire department access routes involves the consideration of many variables, some of which are specified in the requirements in the Code. All these variables should be considered in relation to the type and size of fire department vehicles available in the municipality or area where the building will be constructed. It is appropriate, therefore, that the local fire department be consulted prior to the design and construction of access routes.

NBC(AE) Sentences 8.1.1.1.(2) and (3)

Part 8 Safety Measures at Construction and Demolition Sites
8.1.1.1. Scope
2) This Part applies to fire safety and the protection of the public during the construction, alteration or demolition of every building, including any incompleted or abandoned building.

3) Fire safety at construction and demolition sites shall conform to Section 5.6.
of Division B of the NFC(AE).

NFC(AE) Article 5.6.1.1.

Section 5.6. Construction and Demolition Sites
5.6.1.1. Application (See Note A-5.6.1.1.)
1) This Section applies to fire safety for buildings, parts of buildings, facilities, adjacent buildings or facilities, and associated areas undergoing construction, alteration or demolition operations.

A-5.6.1.1. The degree of application of Section 5.6. should be determined in advance in conjunction with the authority having jurisdiction. In construction, alteration or demolition operations that do not pose an exposure hazard to other buildings or to occupants, the degree of application of Section 5.6. may be minimal.

The degree of application of Section 5.6. to each operation should be determined in advance, as part of the fire safety plan for the operation, taking into consideration such issues as the size of the operation, exposure of adjacent buildings or facilities to hazards, and the site conditions. Operations can range from large multi-storey buildings to small single-storey residences and may include additions or alterations to existing buildings.

NFPA 241, “Standard for Safeguarding Construction, Alteration, and Demolition Operations,” may provide information to assist in the preparation of a fire safety plan as well as guidance for fire protection during construction, alteration and demolition.

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NFC(AE) Article 5.6.1.2.

5.6.1.2. Measures to Mitigate Fire Spread to Adjacent Buildings
1) Measures shall be taken to mitigate fire spread to adjacent buildings and facilities that would be exposed to fire originating from buildings, parts of buildings, facilities and associated areas undergoing construction, alteration or demolition operations. (See Note A-5.6.1.2.(1).)

A-5.6.1.2.(1) Methods and materials used to mitigate fire spread to adjacent buildings and facilities that are deemed necessary following a risk assessment can range from active to passive systems such as spatial separation, installing water curtains, using construction methods and materials that include gypsum sheathing, or erecting a temporary fire barrier such as a fire tarpaulin. Materials that may become part of the finished building must conform to the NBC(AE).

NFC(AE) Article 5.6.1.3.

5.6.1.3. Fire Safety Plan (See Sentence 2.2.1.2.(14) of Division C.)
1) A fire safety plan conforming to Section 2.8. shall be prepared.

NFC(AE) Division C, Sentence 2.2.1.2.(14)

2.2.1.2. Records
14) The fire safety plan required for construction and demolition sites in accordance with Article 2.8.1.1. of Division B shall be accepted in writing by the fire department and the authority having jurisdiction.

NFC(AE) Clause 2.8.1.1.(1)(g)

Section 2.8. Emergency Planning
2.8.1.1. Application
1) Fire emergency procedures conforming to this Section shall be provided for
g) demolition and construction sites regulated under Section 5.6.,

NFC(AE) Article 2.8.2.1.

2.8.2. Fire Safety Plan
2.8.2.1. Measures in a Fire Safety Plan
1) For buildings, parts of buildings and areas described in Article 2.8.1.1., a fire safety plan conforming to this Section shall be prepared in cooperation with the fire department and other applicable regulatory authorities.

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2) The fire safety plan shall be reviewed at intervals not greater than 12 months to ensure that it takes account of changes in the use and other characteristics of the building.

3) The fire safety plan shall include, in addition to the information required in Articles 2.8.2.2. to 2.8.2.11., as applicable, information on

a) the emergency procedures to be carried out in case of fire, including
   i) sounding the fire alarm (see Note A-2.8.2.1.(3)(a)(i)),
   ii) notifying the fire department,
   iii) instructing occupants on the procedures to be followed when the fire alarm sounds,
   iv) evacuating occupants, including special provisions for persons requiring assistance (see Note A-2.8.2.1.(3)(a)(iv)),and
   v) confining, controlling and extinguishing the fire,
b) the appointment and organization of designated supervisory staff to carry out fire safety duties,
c) the training of supervisory staff and other occupants on their responsibilities as regards fire safety,
d) the type, location and operation of the building fire emergency systems, including diagrams,
e) the holding of fire drills,
f) the measures for controlling fire hazards in and around the building, and
g) the inspection and maintenance of building facilities provided for the safety of occupants.
(See Note A-2.8.2.1.(3).)

A-2.8.2.1.(3)(a)(i) These procedures should also include training authorized personnel to silence fire alarm and alert signals under specified conditions. If special keys or devices are required to operate the alarm system, they should be readily available to supervisory staff on duty.

NFC(AE) Article 2.8.2.11.

2.8.2.11. Construction and Demolition Sites

1) Except as required in Sentence (2), prior to the commencement of construction, alteration or demolition operations, a fire safety plan shall be prepared for the site that includes

a) the designation and organization of site personnel to carry out fire safety duties, including a fire watch service if applicable,
b) the emergency procedures to be followed in the event of a fire, including
   i) initiating a fire warning,
   ii) notifying the fire department,
   iii) instructing site personnel on the procedures to be followed once the warning has been initiated, and
   iv) confining, controlling and extinguishing the fire,
c) measures for controlling fire hazards in and around the *building* (see Note A-2.8.2.11.(1)(c)), and
d) a maintenance procedure for firefighting measures required in Section 5.6.

2) Where construction, alteration or demolition operations are carried out in an existing building that is required to have a fire safety plan conforming to this Section, the fire safety plan shall take into account the changes occurring to the building.

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A-2.8.2.11.(1)(c) The control of fire hazards in and around buildings being constructed, renovated or demolished includes fire protection for combustible construction materials and combustible refuse on the site. The sizes of piles of materials and refuse and the location of such piles in relation to adjacent buildings are factors that should be taken into consideration in determining which fire protection measures to implement. The selection of fire protection measures for demolition operations will also depend on the demolition procedure being used, the specific conditions existing on the site, and the firefighting capabilities of the responding fire department.

It is the intent of this Code that requirements regarding the outdoor storage of materials stated in Section 3.3. be referred to and applied at construction and demolition sites.

NFC(AE) Sentences 2.8.2.12.(1) and (3)

2.8.2.12. Retention of Fire Safety Plans
1) The fire safety plan shall be kept in the building for reference by the fire department, supervisory staff, the authority having jurisdiction, and other personnel.
3) The fire safety plan for a *building* or facility within the scope of Sections 3.1., 4.1., and 5.1. shall be kept at the principal entrance to the building or facility.

NFC(AE) Article 2.8.2.13.

2.8.2.13. Distribution
1) A copy of the fire emergency procedures and other duties for supervisory staff, as laid down in the fire safety plan, shall be given to all supervisory staff.

NFC(AE) Article 2.8.2.14.

2.8.2.14. Posting of Fire Emergency Procedures
1) At least one copy of the fire emergency procedures for a building or part of a building shall be prominently posted in each floor area.

2) At least one copy of the fire emergency procedures for an outdoor storage site shall be prominently posted at the outdoor storage site.

3) In every hotel and motel bedroom, the fire safety rules for occupants shall be posted showing the locations of exits and the paths of travel to *exits*.

4) Where a fire alarm system has been installed with no provisions to transmit a signal to the fire department, a sign shall be posted at each manually actuated signalling box requesting that the fire department be notified, and including the telephone number of that department.

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NFC(AE) Article 5.6.1.4.

5.6.1.4. Access for Firefighting

1) Unobstructed access to fire hydrants, portable extinguishers and to fire department connections for standpipe and sprinkler systems shall be maintained.

2) A means shall be provided to allow firefighters to perform their duties on all levels of the building.

3) Provision shall be made for the use of existing elevators, hoists or lifts to assist firefighting personnel in reaching all levels of the building.

4) Access routes for fire department vehicles shall be provided and maintained to construction and demolition sites. (See Note A-5.6.1.4.(4).)

5) Where a construction or demolition site is fenced so as to prevent general entry, provision shall be made for access by fire department equipment and personnel.

A-5.6.1.4.(4) The fire safety plan for the construction or demolition site of buildings conforming to Article 3.2.2.51. or 3.2.2.60. of Division B of the NBC(AE) should take into consideration the design of the access route to the building under construction as well as the elevation of the access route relative to the uppermost floor level to facilitate firefighter access to the roof. (See Sentence 3.2.5.6.(2) of Division B of the NBC(AE).)

NFC(AE) Article 5.6.1.7.

5.6.1.7. Hot Surface Applications

1) Roofing operations and other surface applications that involve heat sources and hot processes shall be considered hot works and shall conform to the requirements in Sentences (2) and (3) and Section 5.2.

2) Bitumen kettles shall

a) not be located on roofs or inside a building,
b) be provided with adequate metal covers that are close-fitting and constructed of steel having a thickness of not less than No. 14 sheet metal gauge thickness,
c) be under constant supervision when in operation, and
d) be maintained free of excessive residue.

3) Mops that have been used for spreading bitumen shall be kept outside the *building* in a safe location when not in use.

Application

This interpretation applies to all construction and demolition sites that range from multi-storey buildings to single-storey residences. Extra considerations should be provided for buildings with irregular configurations or special uses. As this STANDATA does not incorporate all the language from the NBC(AE) and NFC(AE), it is the responsibility of all parties to ensure all other applicable legislation are implemented and adhered to.

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Interpretation

For explanatory purposes, this interpretation provides recommendations for both active and passive systems to mitigate fire spread to adjacent buildings.

Measures to Mitigate Fire Spread to Adjacent Buildings

Most of these suggested solutions are adaptable or scalable and could be used in multiple applications. The degree of application should be determined in advance in conjunction with the SCO/AHJ following a risk assessment. Firefighting capabilities that are available to a municipality in an emergency should also be taken into account when considering protection solutions.

NOTE: NFC(AE) reference Note A-5.6.1.2. provides a non-exhaustive, non-exclusive list of suggestions which notes the following items:

- spatial separation;
- water curtains;
- fire resistive construction methods; or
- temporary fire barriers such as a fire tarpaulin.

Methods that building owners and/or contractors could utilize to provide some of the required protection include, but are not limited to, the following:

- ensure adequate water supply for firefighting operations is provided as soon as combustible construction materials arrive on the site;
- application of an acceptable intumescent or flame-retardant coating in accordance with the manufacturer's directions and the criteria utilized in any performance tests;
- install an acceptable temporary flame/heat-resistant membrane affixed to the exterior sheathing of exposed walls prior to closure and cladding in accordance with the manufacturer's directions and the criteria utilized in any performance tests;
- complete all fire separations, including installation of fire-rated doors, as each floor is completed,
- cover exposed combustible materials as soon as possible;
- provide temporary sprinkler piping with open sprinkler heads that is connected to the standpipe installed on each floor;
- temporarily fence and secure a building site;
- provide on-site security personnel tasked with regular fire watch duties;
- install security lights around the construction area;
- designate certain outdoor areas for smoking;
- limit the amount of combustible and flammable materials stored on site;
- ensure that combustible and flammable materials are properly separated when stored on site;
- ensure that daily clean up and disposal of combustible waste are in secured non-combustible containers;
- provide a fire watch at the beginning of every break period on site in addition to a comprehensive site inspection at the end of each day;
- install temporary electronic intrusion and thermal detection systems; or
- plan and coordinate non-sequential construction in new residential subdivisions.

These methods are only some of the possible solutions to mitigate fire spread. Some may only be required for a specific duration when on site conditions or activities create increased vulnerability.

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There are other provisions in Section 5.6 that should be considered if applicable to the construction site. These include procedures and permits required for hot works, when and how to properly refuel and use fuel-fired appliances and equipment, and how to properly use temporary heating during construction

The NBC(AE) provides required limiting distances, spatial separations, and the maximum permitted unprotected openings for exposed walls as a completed assembly at the conclusion of construction. When two adjacent structures are expected to be at the framing or sheathing stage at the same time is when these structures are most vulnerable to flame spread, and owners and/or contractors should consult with the SCO/AHJ to determine the appropriate measures are taken.

Innovation, collaboration, and discussion between contractors, owners, developers, local SCOs (building and fire) and Occupational Health and Safety Officers professionals should result in additional solutions that meet the need for protection and determining when the use of each of the identified solutions, singly or in sets, would be the most effective for the owner and acceptable to the SCO/AHJ.

Although this document is general in scope for measures to mitigate fire spread to adjacent buildings the NFC(AE) contains specific requirements for residential and business occupancies of combustible construction or encapsulated mass timber construction up to six and 12 storeys. These requirements should be considered and included if applicable.

Water Supplies and Access for Firefighting

The owner, developer, or contractor responsible for a construction or demolition site is required to incorporate the following measures:

- provide the fire department with unobstructed access to roads, routes, stairways, and lifts (where applicable) that accommodate firefighting personnel, apparatus, and equipment;
- ensure that unobstructed access is always provided to fire hydrants (or other water supplies and connections serving the site that is acceptable to the SCO/AHJ), portable extinguishers, and fire department connections for standpipe and sprinkler systems on the site; and
- verify that water supply systems, where required by the NBC(AE), the NFC(AE), municipal infrastructure standards, development permit conditions, and/or subdivision approvals, are operational and sufficient for firefighting operations.

It is critical that regular communication between the owner(s), developer(s), municipality, and the building and fire authorities occur to ensure that these measures and equipment are maintained and continue to operate to an acceptable level throughout the duration of construction, demolition, or alteration where applicable.

Given the potential complexity that may be involved in undertaking these measures on any particular site(s), the following should be taken into consideration for water supply and access for firefighting:

- weather;
- location;
- road surfaces;
- geography;
- topography;
- security provisions;
- shared access to infrastructure;
- impacts of other related activities;

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- material storage;
- parking of vehicles and equipment;
- multiple employees and/or employers;
- multiple owners of multiple sites; and
- other issues that may arise during construction or demolition.

All the measures noted above are required to be taken under consideration and addressed where applicable in the required fire safety plan as per Article 5.6.1.3. and Section 2.8. of the NFC(AE).

This STANDATA replaces 19-FCI-005/19-BCI-016, “Protection of adjacent buildings during construction and demolition.”

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition and the National Fire Code – 2023 Alberta Edition

Issued by the Provincial Building and Fire Administrators

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